

Ashish Goswami

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EDUCATION

IIT-Delhi

PhD, School of AI
2022-2026 (Tentative)
Cum. GPA: 8.8/ 10.0

IIIT-Guwahati

B.Tech in Electronics and Comm.
2018-2022
Cum. GPA: 8.5/ 10.0

SKILLS

Technical Skills:

Deep Learning, Computer Vision,
NLP, Adversarial Networks

Frameworks:

Pytorch, Lightning, Keras, OpenCV,
Flask

Languages and Libraries:

Python, C, Java, Numpy, Pandas

Tools:

Git, Vim, AWS

COURSEWORK

Machine Learning
Deep Learning
Advance Computer Vision
Mathematics for MINDS
Data Structure and Algo.
Linear Algebra
Calculus and Probability
Computer Architecture
Signal Processing

ACTIVITIES

Mavericks: The ML Society
Co-ordinator
IIIT-G

01.19.2025

EXPERIENCE

Delhivery

Associate Data Scientist

Nov 2021 – Aug 2022

Gurugram, IN

- Worked on improving serviceability polygon coverage to 99.9% of shipments utilizing Peta-Byte scale data.

Applied Data Science Intern

Gurugram, IN

- Worked on Computer Vision based applications to optimize truck load and empty trolley utilization at Sorting facilities, improving mAP by 7%.
- Worked on improving capabilities of spell correction module using DL.

Holosuit Pte. Ltd.

Computer Vision Intern

June 2021 – Nov 2021

Mysore, IN

- Contributed to building scalable Knowledge Graphs based closed domain conversational AI as a QA system
- Worked on Python and Unity based Robotics framework for teaching assistance.

Spyne.ai

Computer Vision Intern

Aug 2020 - Nov 2020

Gurugram, IN

- Worked on High Resolution background matting model for various e-commerce categories.
- Built Auto Image Tagging and Title generation models for fashion clothing items using Tensorflow, Flask, AWS for scalable deployment.

SJTech Solutions Pvt. Ltd.

Machine Learning Development Intern

May 2020 - July 2020

Bhopal, IN

- Developed an end-to-end ML system similar to AutoML for tabular data.
- Worked on retail chatbots, as well as GAN based face image generation tools.

RESEARCH PUBLICATIONS

[1] Harman Singh, Poorva Garg, Mohit Gupta, Kevin Shah, Ashish Goswami, Satyam Kumar Modi, Arnab Kumar Mondal, Dinesh Khandelwal, Parag Singla, Dinesh Garg

"Image Manipulation via Multi-Hop Instructions - A New Dataset and Weakly-Supervised Neuro-Symbolic Approach" At EMNLP 23-Main

[link]

ACHIEVEMENTS

Received TCS Research Fellowship

Kaggle Expert (Rank 420/40,888)
[Profile Here]

Smart India Hackathon 2020
National Level Finalist for
DL based Road Damage
Detection solution. [Code]

Secured 140th rank out of 10,000+
participants on Amazon ML
Challenge on Hackerearth.

Won the Attentive.ai's Image
Classification Competition hosted
on Dockship.io. [Code here]

Finalist at National
Astronomy Olympiad.

REFERENCES

Prof. Parag Singla
Professor IIT-Delhi (CSE, HEAD School
of AI)
email: parags@cse.iitd.ac.in

Prof. Prathosh AP
Asst. Professor IISc. (Dept. of
Electronics and Communication)
email: prathosh@iisc.ac.in

Dr. Shovan Barma
Asst. Professor IIIT-G (Dept. of
Electronics and Communication)
email: shovan.barma@gmail.com

[2] Ashish Goswami, Satyam Kumar Modi, Santhosh Rishi Deshineni, Harman Singh, Prathosh A. P, Parag Singla

"GraPE: A Generate-Plan-Edit Framework for Compositional T2I Synthesis" Arxiv Preprint

[link]

PROJECTS

Leukemia Detection from Blood Cell Images

Leukimia detection have very imbalanced dataset, overcame this by using Cross-validation and penalizing dominant class.

Trained Efficient B3 with custom standardization and LR schedule to obtain 94.5 f1 score on TPUs.

Multi-Modal Book cover classification

This was part of my final assignment for COL775@IIT-Delhi. Given image, title pairs we had to correctly classify the book genre. Using a fine-tuned CLIP for multi-modal representation, I was able to rank #1 in a class of 120 students

Image Captioning App

Incorporating CNNs and Transformers to create a lightweight Image Captioning model to work on Android using Flutter and Tensorflow.

Resilience Analysis of GANs and introduction of approximations

Analysed the error resilience shown by GANs (DCGANs, WGANs, and CycleGANs) so as to employ approximations in them

Hand Written Text Recognition

2nd Prize, Rethink UX's text Recognition Challenge, 2020

Used OpenCV pipeline to segment lines from a page.

Utilised an Encoder(CNN)-Decoder(Transformer) architecture to achieve 94% acc.

Task Automation using Gesture Recognition

Manually collected data using OpenCV and webcam (1000/ category)

Trained CNN model with attention to attain an accuracy of over 99%.

Several Repetitive tasks like voice assistant(self-made), Chrome, Vol Up/Down etc.were mapped to the categories

Friday

A voice assistant made from scratch to perform several day-to-day tasks, later integrated with the Gesture Recognition Project